you. That person sitting across the café from you is, from your perspective, several nanoseconds in the past, which may go part of the way toward explaining their wistful expression and outdated fashion sense. Everything you see is in the past, as far as you're concerned. If you look up at the Moon, you're seeing a little over a second ago. The Sun is more than eight minutes in the past. And the stars you see in the night sky are deep in the past, from just a few years to millennia.

The concept of this kind of light speed delay might already be familiar to you, but its implications are profound. It means that as astronomers, we can look into the sky and watch the evolution of the universe happen, from its early beginnings to the present day. We use the unit "light-year" in astronomy not just because it's conveniently huge (about 9.5 trillion kilometers, or 5.9 trillion miles), but also because it tells us how long light has been traveling from the thing we're looking at. A star 10 light-years away is 10 years in the past, from our perspective. A galaxy 10 billion light-years away is 10 billion years in the past. Since the universe is only about 13.8 billion years old, that 10-billion-light-year-distant galaxy can tell us about the conditions of our universe when it was still in its youth. In that sense, looking out into the cosmos is tantamount to looking into our own past.

There's an important caveat to this, and I would be remiss if I didn't mention it. We technically can't see our *own* past at all. The light speed delay means that the more distant a thing is, the farther in the past it is, and that relationship is strict: not only can we not see our own past, we can't see those distant galaxies in the present, either. The more distant something is, the farther away it is on a timeline of the cosmos.

So how do we learn anything useful about our own past if we're only seeing the past for some other galaxy, long ago and far away? It comes down to a principle so central to cosmology that it is literally called the *cosmological principle*. Simply stated, it's the idea that for all practical purposes, the universe is basically the same everywhere. Obviously this isn't true on human scales—the surface of the Earth is pretty importantly different from deep space or the center of the Sun—but on the kind of astronomically large scales on which whole galaxies can be tallied up as individually uninteresting specks, the universe looks the same in every direction, and is made of all the same stuff.[fn1] This idea is closely related to the Copernican Principle, which is the erstwhile heretical notion stated by